Recognising special number sequences

Task A: Arithmetic and geometric sequences

1) Sort these sequences into the correct column of the table.

arithmetic	geometric	neither
a) 1, 2, 3, 4, 5,…	b) 3, 9, 27, 81,…	d) 1, 1, 2, 4, 7,…
g) −1, 5, 11, 17, 23,.	c) 0.5, 1, 2, 4,…	e) 1, 2, 4, 7, 11,.
i) 36, 12, −12, −36,.	f) −2, 4, −8, 16,..	h) 18, 35, 52, 68,.
	j) 13.5, 9, 6, 4, …	k) 12, 6 , 3 , 0, …

a) 1, 2, 3, 4, 5, …
b) 3, 9, 27, 81, ….
c) 0.5, 1, 2, 4, …
d) 1, 1, 2, 4, 7, …
e) 1, 2, 4, 7, 11, …
f) −2, 4, −8, 16, …
g) −1, 5, 11, 17, 23, …
h) 18, 35, 52, 68, …
i) 36, 12, −12, −36 …
j) 13.5, 9, 6, 4, …
k) 12, 6 , 3 , 0, …

2) Sort these sequences into the correct column of the table.

arithmetic	geometric	neither
a) $\frac{3}{2}, 1, \frac{1}{2}, 0, ..$	b) $\frac{3}{8}, \frac{3}{4}, \frac{3}{2}, 3, ..$	d) $\frac{3}{5}, \frac{3}{10}, \frac{3}{15}, \frac{3}{20}, ..$
g) $-\frac{4}{5}, -\frac{2}{3}, -\frac{8}{15}, -\frac{2}{5}, ..$	c) $\frac{1}{6}, \frac{5}{3}, \frac{50}{3}, \frac{500}{3}, ..$	f) $-\frac{8}{5}, 1, \frac{18}{5}, 6, ..$
	e) $-3, 1, -\frac{1}{3}, \frac{1}{9}, ..$	

a) $\frac{3}{2}, \ 1, \ \frac{1}{2}, \ 0, \ \ldots$
b) $\frac{3}{8}, \ \frac{3}{4}, \ \frac{3}{2}, \ 3, \ \ldots$
c) $\frac{1}{6}, \ \frac{5}{3}, \ \frac{50}{3}, \ \frac{500}{3}, \ \ldots$
d) $\frac{3}{5}, \ \frac{3}{10}, \ \frac{3}{15}, \ \frac{3}{20}, \ \ldots$
e) $-3, \ 1, \ -\frac{1}{3}, \ \frac{1}{9}, \ \ldots$
f) $-\frac{8}{5}, \ 1, \ \frac{18}{5}, \ 6, \ \ldots$
g) $-\frac{4}{5}, \ -\frac{2}{3}, \ -\frac{8}{15}, \ -\frac{2}{5}, \ \ldots$

3) A sequence starts $2a, 6a, \ldots$

a) If the sequence was arithmetic, what would the next three terms be?

$10a, 14a, 18a, \ldots.$

b) If the sequence was geometric, what would the next three terms be?

$18a, 54a, 162a, \ldots$

c) Continue this sequence in another way, describe the rule you have used.

Many possible answers.

Task B: Exploring other sequences

1) By working out the pattern, find the next two terms in each sequence.

a) 2, 8, 32, 128, … *512, 2048* *Geometric sequence with common ratio ×2*

b) 2, 8, 14, 20, … *26, 32,* *Arithmetic sequence with common difference +6*

c) 2, 8, 15, 23, … *32, 42* *Quadratic sequence with second difference +1*

d) 2, 8, 16, 26, … *38, 52* *Quadratic sequence with second difference +2*

e) 2, 8, 10, 18, … *28, 46* *Add the two previous terms to get the next term*

f) 2, 8, 17, 29, … *44, 62* *Quadratic sequence with second difference +3*

g) 8, 2, 10, 12, … *22, 34* *Add the two previous terms to get the next term*

h) 8, 2, −4, −10, … *−16, −22* *Linear sequence with common difference −6*

i) 3, 5, 11, 21, … *35, 53* *Quadratic sequence with second difference +4*

j) 4, 6, 12, 22, … *36, 54* *Quadratic sequence with second difference +4*

k) 6, 10, 22, 42, … *70, 106* *Quadratic sequence with second difference +8*

l) −3, −5, −8, −13, … *−21, −34* *Add two previous terms to get the next term*

m) −3, −5, −7, −9, … *−11, −13* *Arithmetic sequence with common difference −2*

n) 2, −6, 18, −54, … *162, 486* *Geometric sequence with common ratio ×(−3)*

o) 1, −7, −10, −8, … *−1 11* *Quadratic sequence with second difference + 5*

−8 −3 +2 +7 +12

+5 +5 +5 +5

2) Aisha, Izzy and Lucas have all created a sequence starting with the terms:

$\frac{1}{6}, \frac{1}{2}, \ldots$

For each question justify your answer.

$\frac{1}{6}, \frac{1}{2}, 1\frac{1}{6}, 2\frac{1}{6}, \ldots$

Aisha

$\frac{1}{6}, \frac{1}{2}, \frac{5}{6}, 1\frac{1}{6}, \ldots$

Izzy

$\frac{1}{6}, \frac{1}{2}, 1\frac{1}{2}, 4\frac{1}{2}, \ldots$

Lucas

a) Who has created an arithmetic sequence?

Izzy, common difference $+\frac{1}{3}$

b) Who has created a geometric sequence?

Lucas, common ratio × 3

c) Who has created a quadratic sequence?

Aisha, the sequence $+\frac{1}{3}$ *then* $+\frac{2}{3}$ *then* $+\frac{3}{3}$

common second difference of $+\frac{1}{3}$